

Virum Ranka

Software Engineer | Large-Scale Data & ML Systems | MS Data Science, U-Michigan (Dec 2026)
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SUMMARY

Engineer with 2+ years designing, building, testing, and deploying production software end to end: distributed data pipelines at 1,800+ pipeline scale, event-driven services on AWS and Azure, and multi-agent LLM systems built for on-call engineering teams. B.Tech CS foundation in algorithms, databases, and distributed systems; MS Data Science at Michigan.

EXPERIENCE

AI Data Scientist | *DTE Energy, Internship* Jun 2026 – Present

- Cut incident diagnosis 75% (20→5 min) across 1,800+ production Azure Data Factory pipelines by designing and shipping a multi-agent observability service with a three-layer hybrid retrieval strategy over logs, metadata, and runbooks.
- Build the agent runtime end to end (human-in-the-loop middleware, a virtual filesystem for agent state, tool APIs, skill injection, and deterministic execution paths), with context management that keeps long-running sessions stable.
- Sustain an 85% first-action success rate on mitigation recommendations: 17 of 20 monthly incidents matched the agent's diagnosis within 24 hours of on-call response.
- Instrument every reasoning step (tool calls, latency, token count, confidence decision) into a queryable MLflow store, enabling weekly regression evaluations against 30+ golden incident cases.

Programmer | *Michigan Medicine, U-M Medical School* Apr 2026 – Jun 2026

- Built an AI research and writing assistant for medical faculty, orchestrating multiple agents with LLM chaining and retrieval-augmented generation to automate literature synthesis, NIH grant drafting, and author-style adaptation.
- Designed modular agent pipelines around structured prompt engineering, task planning, and typed data contracts, then hardened them with regression testing on pipeline reliability and output quality across diverse biomedical tasks.
- Debugged multi-step agent failures and validated LLM outputs iteratively, refining agent behavior through systematic feedback cycles.

Research Assistant | *College of LSA, University of Michigan* Nov 2025 – Present

- Developed DuRAG, a dual-layer graph RAG system that builds knowledge graphs from unstructured research data spanning 353+ variables; multi-agent orchestration improved task accuracy 35%.
- Built LLM evaluation benchmarks with ground-truth datasets and few-shot prompting for structured-output validation across 4 baseline models (Gemini 3, Claude 4.5, ChatGPT, DataSci Bench).
- Ran statistical and ML analysis of constitutive skin-pigmentation variation for the SAPPHIRE genetics initiative.

AI/ML Engineer | *Spect Pvt Ltd, Internship* Aug 2024 – Dec 2024

- Fine-tuned BERT/RoBERTa with weighted cross-entropy loss on a 25K-sample corpus, lifting macro-averaged recall 18% for multi-class intent classification.
- Automated daily ingestion of 10K+ text samples through event-driven AWS services (EventBridge, Lambda, SageMaker, DynamoDB), removing ~40% of manual preparation work.

PROJECTS

GPT-style Language Model & Conversational SFT GitHub

- Implemented a GPT-style transformer from scratch in PyTorch (multi-head causal self-attention, RoPE, SwiGLU FFN, RMSNorm), with a mixed-precision, gradient-accumulating pretraining pipeline over FineWeb-Edu.
- Added a conversational SFT stage using selective loss masking on non-assistant tokens to align instruction-following behavior.

Multi-Agent Campaign Concept Generator GitHub · Live

- Shipped a deployed LangGraph system of specialized agents (research, strategy, creative, validation) with ChromaDB retrieval and LLM quality gates for output validation.

TECHNICAL SKILLS

Languages: Python, Java, SQL, R, Shell Scripting
Systems & Cloud: AWS (Lambda, EventBridge, SageMaker, DynamoDB, EC2), Azure Cloud, Docker, Databricks (DLT, Delta Lake, Vector Search), Apache Kafka
Data: PostgreSQL, MongoDB, Snowflake, Pandas, NumPy
ML & AI: PyTorch, Hugging Face Transformers, TensorFlow, Scikit-learn, LangChain/LangGraph, RAG, MLflow, OpenCV
Practice: Git/GitHub, CI/CD, regression & A/B testing, Agile/Scrum · **Certifications:** AWS Cloud Foundations, AWS Data Engineering

EDUCATION

MS Data Science, University of Michigan, Ann Arbor Aug 2025 – Dec 2026

Coursework: Applied Machine Learning, Natural Language Processing, Web Systems, Database Design, Statistics

B.Tech Computer Science, University of Mumbai | GPA 9.2/10 Aug 2021 – May 2025

Coursework: Data Structures & Algorithms, Database Management Systems, Cloud Computing, Big Data Engineering, Software Engineering

ACHIEVEMENTS & LEADERSHIP

- 2nd Place, Software category, TechExpo 2024, selected from 200+ competing projects.
- Led a 24-person team organizing DataHack 2.0, running the event for 350+ participants nationwide.